

ECOCIDE

A Satellite-Based Evidentiary Framework for War-Time Environmental Crimes

Maps and Plots — Complete Figure Set

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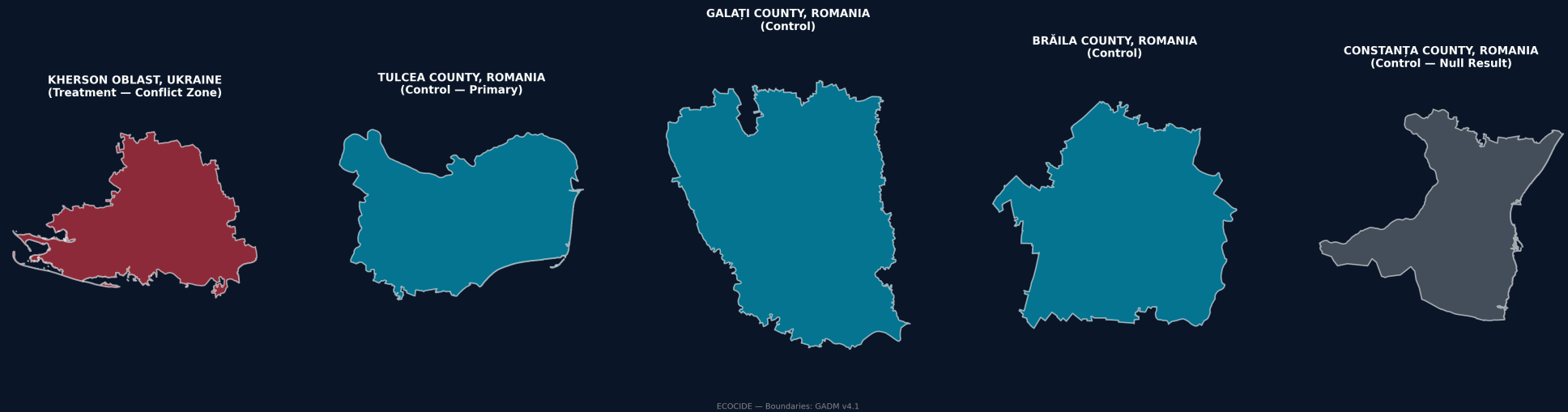
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Figure 1 — Study Area Overview

ECOCIDE — STUDY AREA

Difference-in-Differences Design: Kakhovka Dam Destruction (6 June 2023) — 1 treatment zone, 4-control panel



Treatment zone (Kherson Oblast, Ukraine) and the four-control panel (Tulcea, Galati, Braila, Constanta counties, Romania) used in the Difference-in-Differences design. Boundaries: GADM v4.1.

Figure 2 — Before the Dam's Destruction (May 2023)

BEFORE (MAY 2023)



0

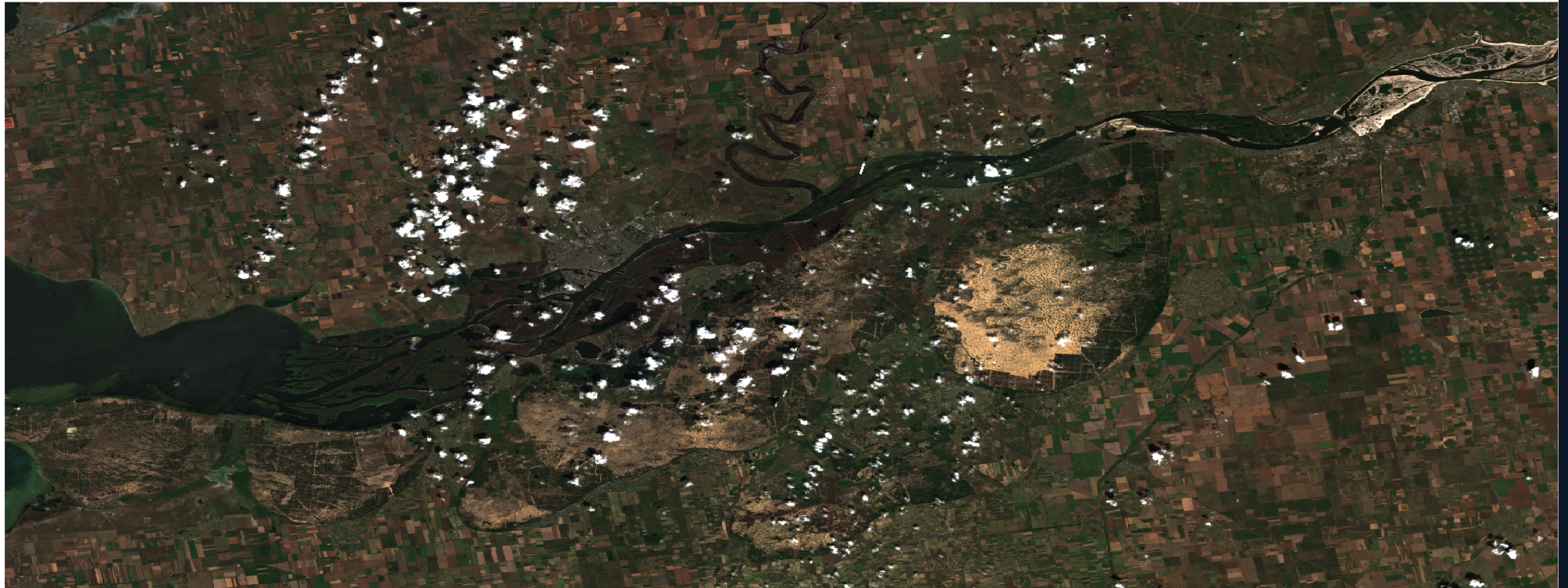
30

60 km

Sentinel-2 true-colour imagery of the Kakhovka reservoir immediately before the dam's destruction, showing the full reservoir.

Figure 2 — After the Dam's Destruction (July 2023)

AFTER (JULY 2023)



0

30

60 km

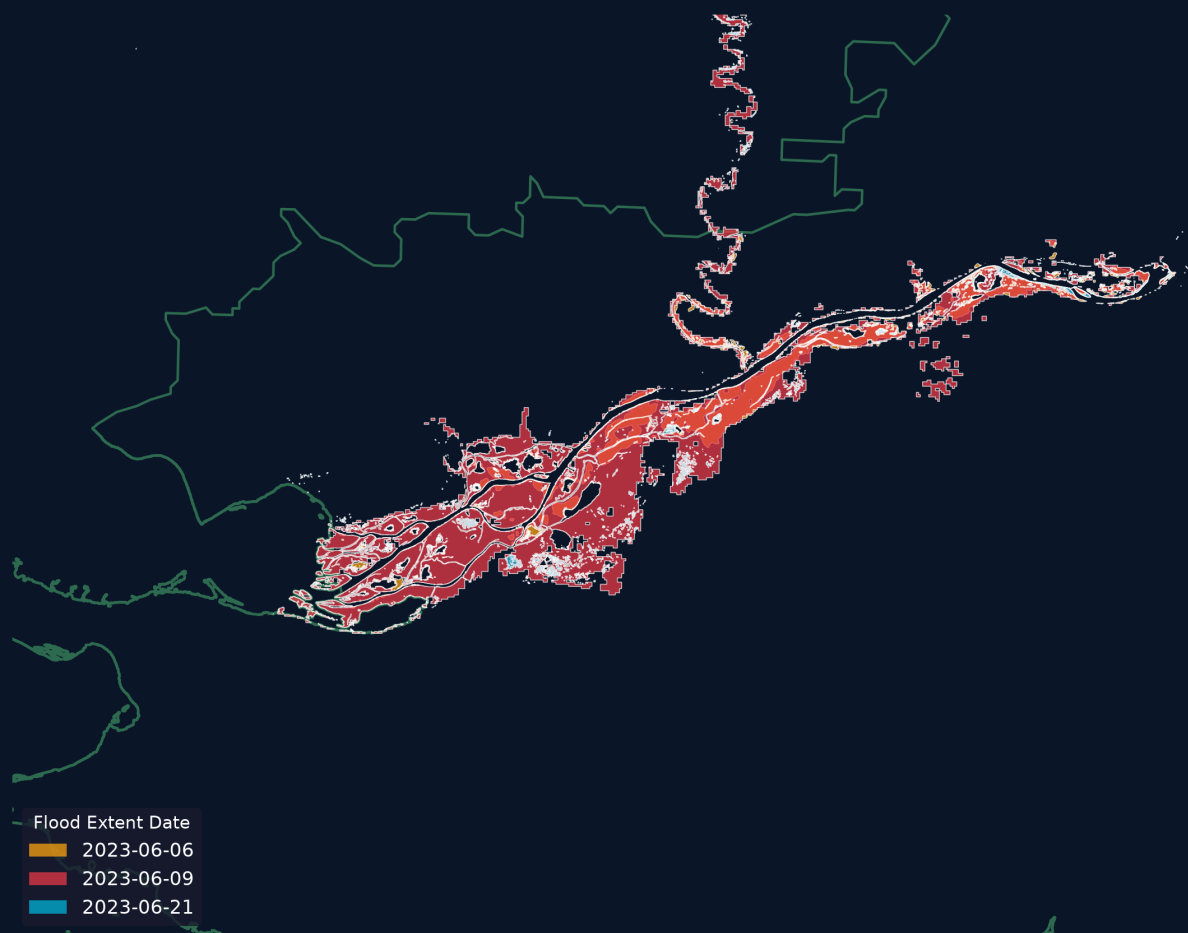


Sentinel-2 true-colour imagery of the same area after the breach, showing near-complete drainage of the reservoir and exposure of the former lakebed.

Figure 3 — Verified Flood-Extent Map

KAKHOVKA DAM FLOOD EXTENT

Verified multi-sensor flood polygons, Kherson Oblast, Ukraine

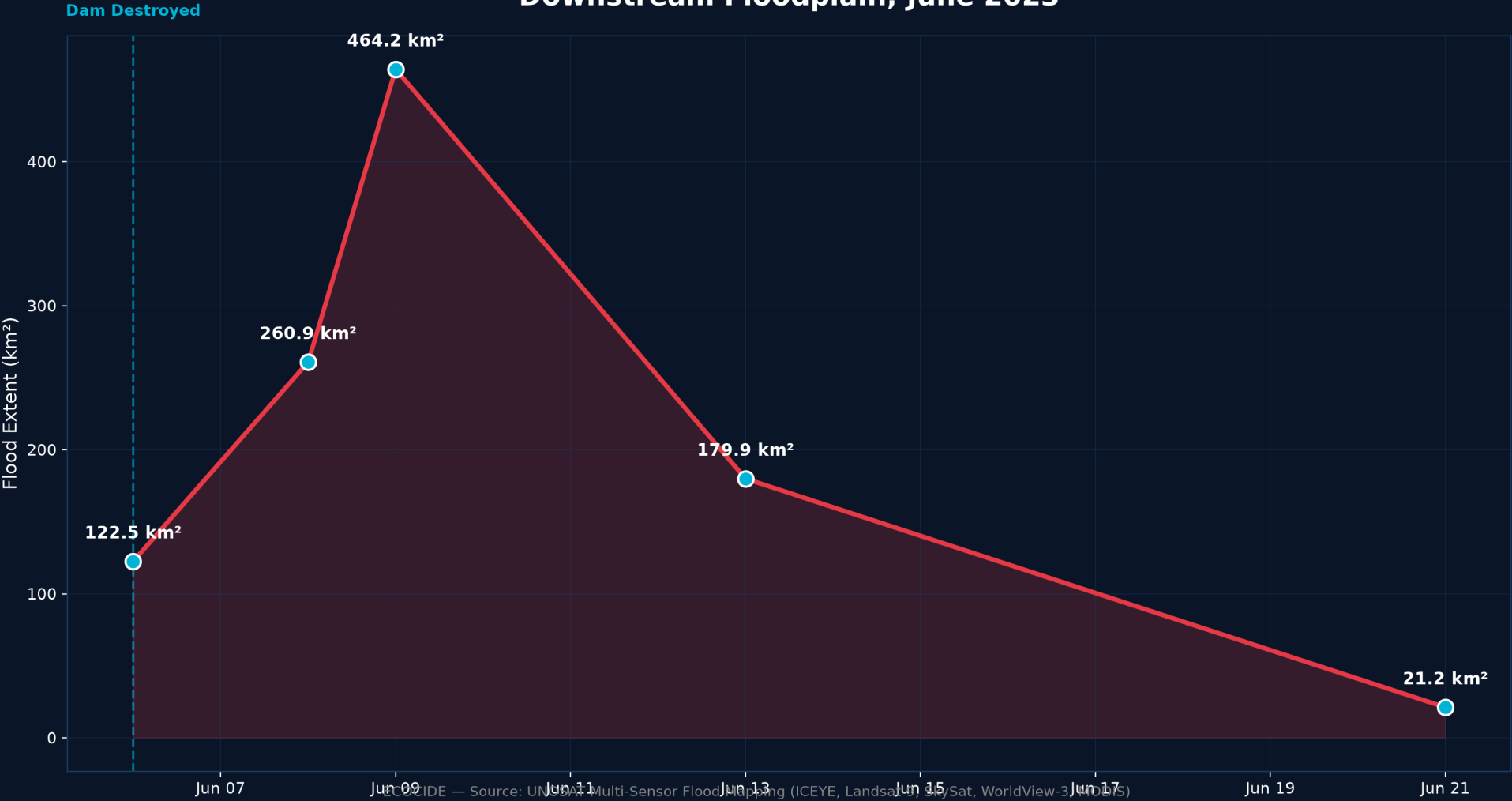


ECOCIDE — Source: UNOSAT Multi-Sensor Flood Mapping

UNOSAT multi-sensor flood-extent polygons over the Kherson Oblast flood corridor at three dates (6, 9, 21 June 2023), showing the flood's rise, peak, and recession.

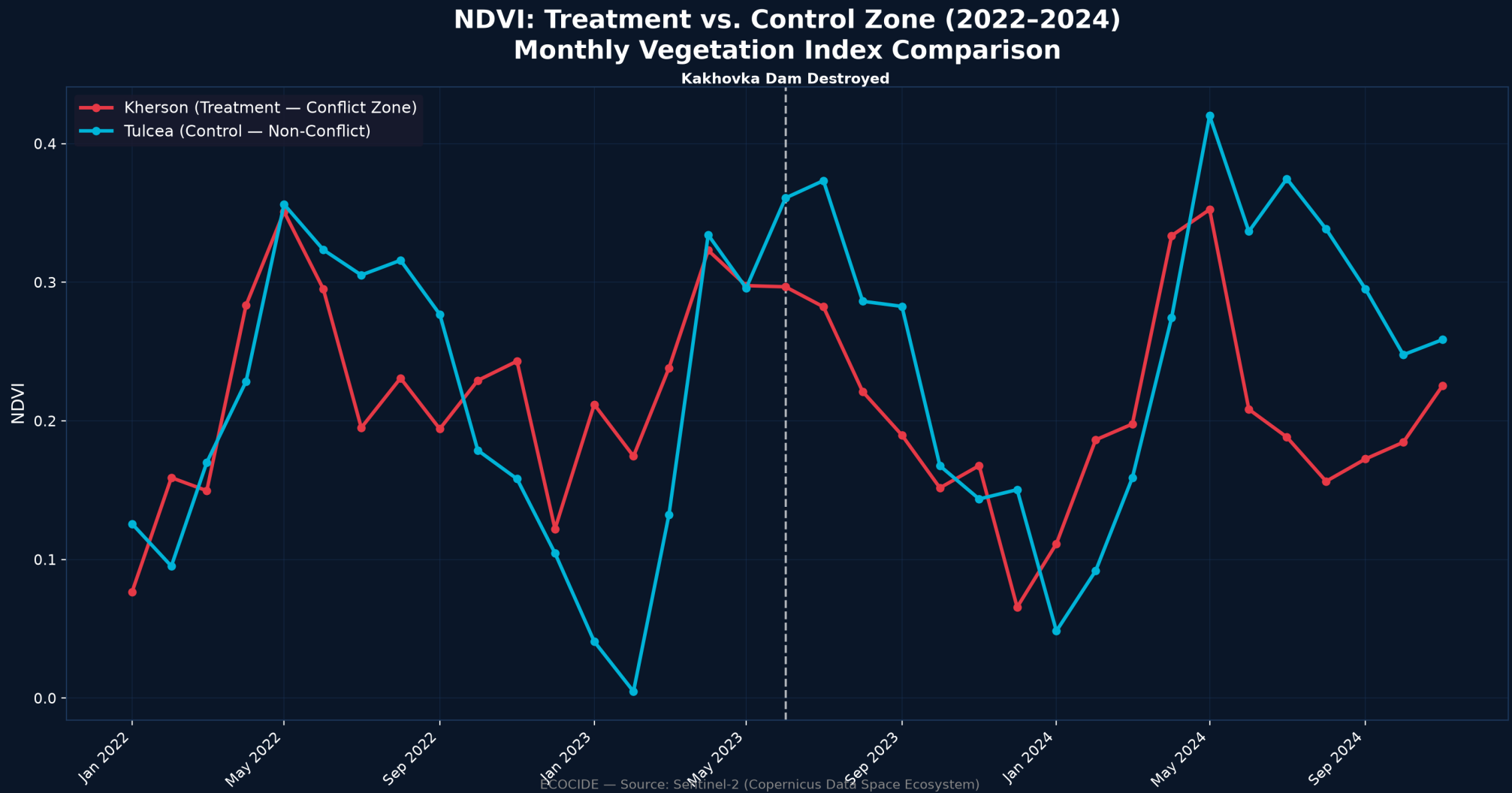
Figure 4 — Flood Hydrograph

Kakhovka Dam Flood Extent Progression
Downstream Floodplain, June 2023



Verified flood extent over time: 122.50 km² (6 June) → 464.18 km² peak (9 June) → 21.17 km² (21 June) — a complete rise-peak-recession cycle within two weeks.

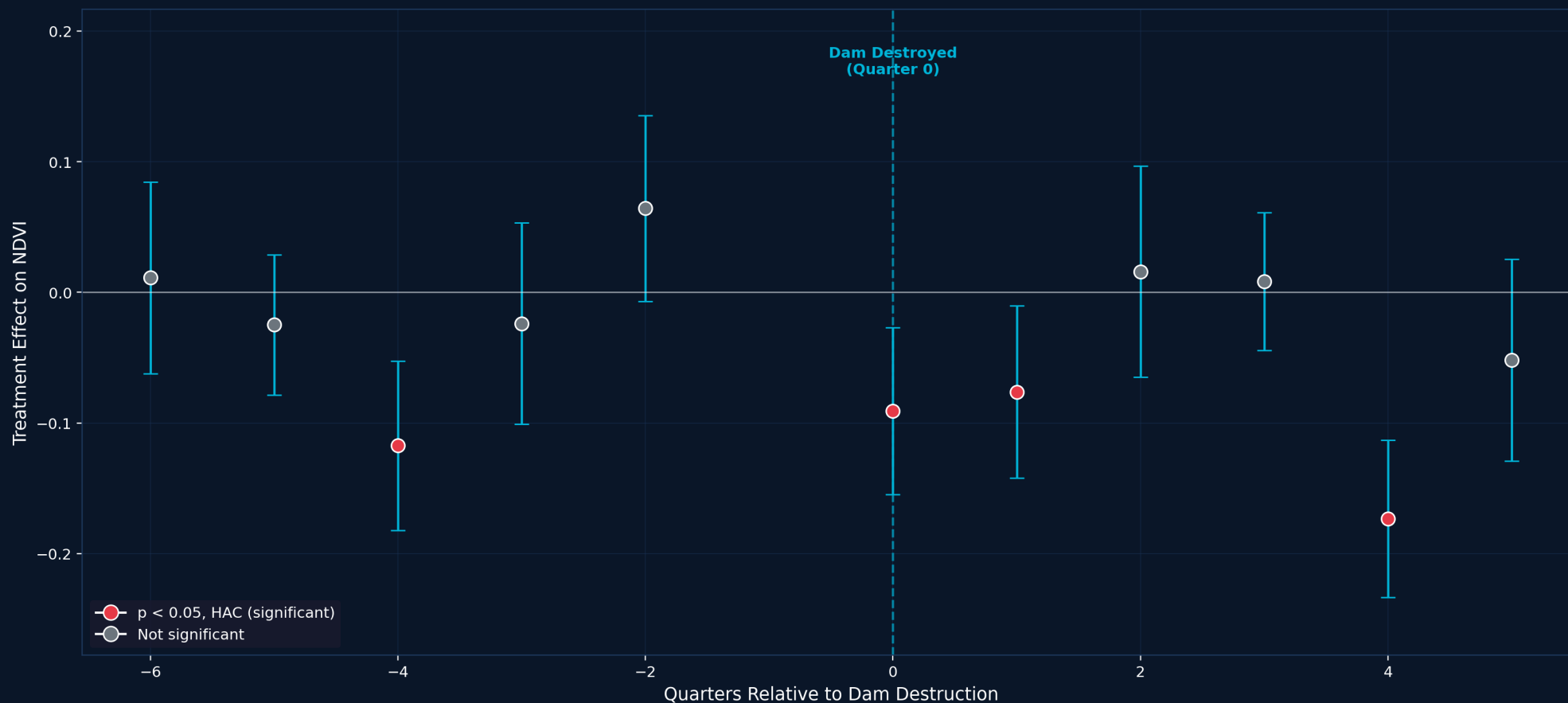
Figure 5 — NDVI: Treatment vs. Control Over Time



Monthly mean NDVI for Kherson (treatment) and Tulcea (control), Jan 2022 – Dec 2024, showing a visible divergence after the dam's destruction.

Figure 6 — Quarterly Event Study

**Event Study: Quarterly Treatment Effect on NDVI
Relative to Kakhovka Dam Destruction (June 2023)**

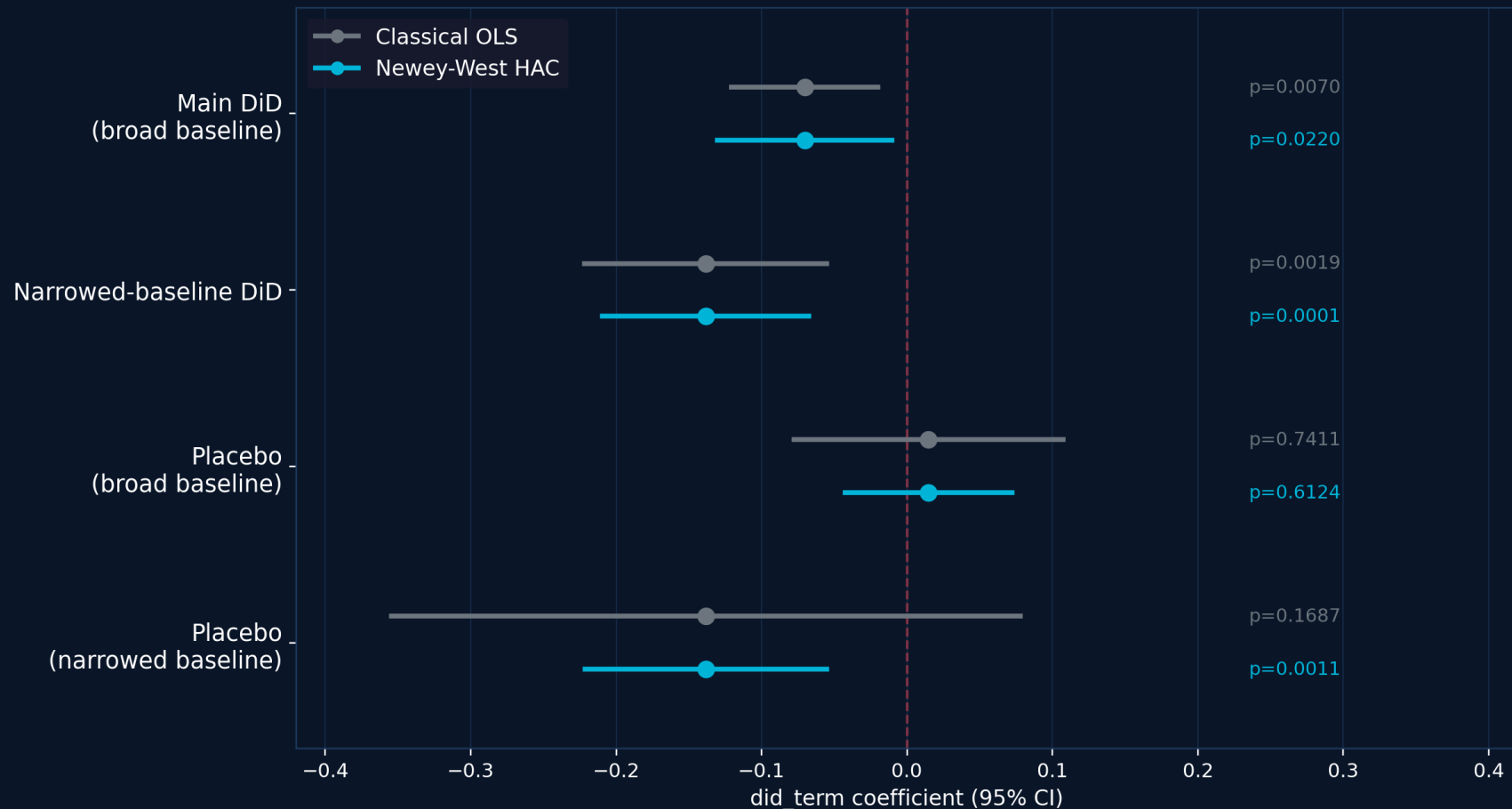


ECOCIDE — Quarterly-binned event study, month fixed effects, Newey-West HAC standard errors
Quarterly treatment-effect estimates (Newey-West HAC standard errors) relative to the event date,
including the disclosed pre-treatment-quarter signal.

Figure 7 — Robustness: Classical vs. HAC Standard Errors

ROBUSTNESS: CLASSICAL vs. HAC STANDARD ERRORS

Point estimates and 95% confidence intervals across all four causal-inference models



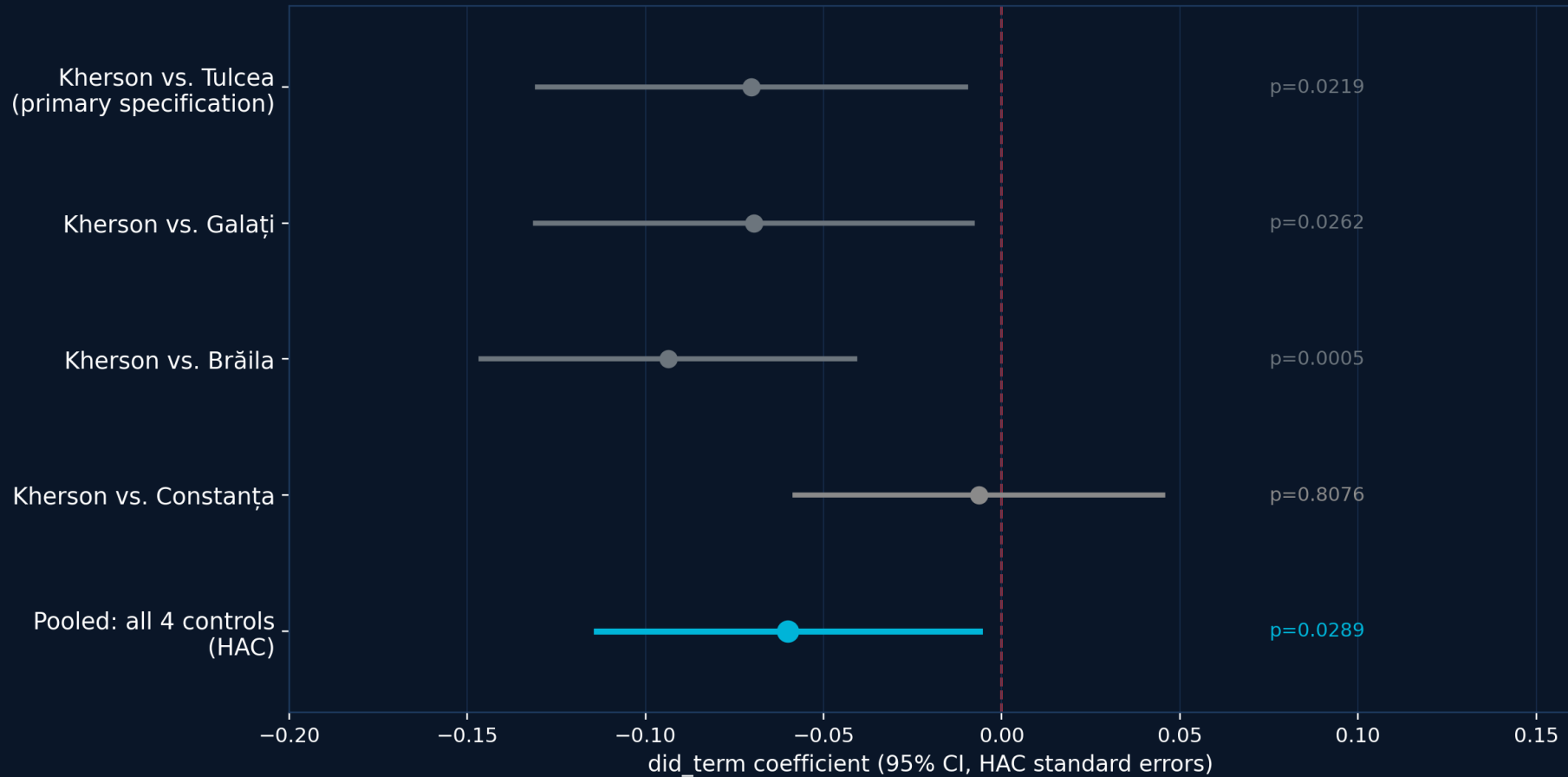
ECOCIDE — did_model.py / did_model_narrowed.py / placebo_test.py / placebo_narrowed.py

Point estimates and 95% confidence intervals for all four causal-inference models under classical OLS versus Newey-West HAC standard errors.

Figure 8 — Multi-Control Robustness Check

MULTI-CONTROL ROBUSTNESS CHECK: DOES THE EFFECT HOLD AGAINST EACH CONTROL?

Kherson vs. each of 4 Danube/Black Sea control counties individually, plus the pooled 4-control panel



ECOCIDE — did_model_multi_control.py

Kherson's NDVI decline tested against each of four control counties individually, plus the pooled four-control estimate. Three of four controls reproduce a significant effect; Constanța does not.